

Deep Learning-Based Small Fish Species Classification Using Transfer Learning and Grad-CAM Explainability

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Abstract

Here we present an extensive image dataset for ten native Bangladesh small fishes termed as Bele (*Glossogobius giuris*), Chanda Nama (Chanda nama), Chela (*Salmostoma bacaila*), Guchi (*Mastacembelus pancalus*), Kachki (*Corica soborna*), Mola (*Amblypharyngodon mola*), Kata Phasa (*Stolephorus*), Pabda (*Ompok pabda*), Puti (*Puntius sophore*), and Tengra (*Mystus vittatus*). Introduction: This dataset was specifically developed to help in the study and research of fish species identification, classification and biodiversity monitoring. Sister Species were obtained from different fish markets in the capital Dhaka city. Many varieties of fish are sent to Dhaka city from different geographic localities of Bangladesh. Altogether, this dataset guarantees a representative sampling of local aquatic diversity.

Images were taken with a smartphone camera under the same controlled and constant conditions to keep the same settings of the camera and ensure uniformity between samples. All specimens were viewed against a neutral background with uniform lighting conditions. By reducing variability in the environment and improving image quality for analytical purposes. Original high-resolution images are included in the dataset, and they were augmented with standard approaches. This augmentation included rotations, flipping and brightness. By doing this, the dataset is expanded, and it is more useful for training robust machine learning (ML) and deep learning (DL) models for computer vision.

This dataset can be reused in many different areas. This dataset will be an essential guide for researchers and industrial people for building automatic fish species Identifying and classifying systems, especially in the case of the Bangladesh aquaculture diversity. In addition, by enabling the tracking of native fish species distribution and population behaviour, the dataset can also assist with ecological and environmental studies and research. Having this in a reusable structured format may enable integration into ML/DL pipelines which might help lay the foundations in areas such as fisheries management, sustainable aquaculture, conservation biology, and studies in economics and culture. This dataset is therefore incredibly important to the process of combining technological progress and environmental preservation. We hereby document how the data can be utilized, the structure of the dataset, method of data collection and augmentation methods to ensure transparency and reproducibility for further research.

1 Introduction

Bangladesh has a wide range of freshwater fish species where many of them are small indigenous fish (SIS). Species including Puti, Tengra, Mola, Kachki and Chanda are vital for ecological balance, nutritional food supply and local economy. As per high resemblance between fishes at species level, lighting variations, background noise and inconsistency in the process of image capture, addressing the challenge of image recognition is still a major concern. Identification of fish species based on traditional taxonomic approaches requires access to expert knowledge and is time-consuming, and thus limits its efficiency for large-scale fish monitoring, fish market studies, and biodiversity assessments [1, 3, 3].

Using advancement of computer vision, deep based-learning approaches, in particularly Convolutional Neural Network (CNN), showed high performance for image based classification task. Transfer learning enable the use of accurate classification even when you have little or no training since these methods often use image representation extracted by models pre-trained on a large scale dataset. As such, common models include MobileNetV2, ResNet50V2, and EfficientNetB0 which are used for their effectiveness, generalization, and strength overall.

Introduction: Classification of fish images has made significant advancement, but few studies have addressed small fish species of Bangladesh because there are not enough public datasets, the fish images are heterogeneous with other objects and backgrounds, and the model decisions are not always explainable. In order to solve these problems, this work proposes an entire end-to-end deep learning framework for small fish species classification using SmallFishBD dataset. In order to increase the quality and robustness of the data for the study, a comprehensive preprocessing pipeline is applied, consisting of resizing, brightness correction, denoising, Gaussian blur, rotation, and augmentation.

The proposed approach standardizes the metrics used to evaluate three transfer learning models (MobileNetV2, ResNet50V2, EfficientNetB0) across accuracy,

loss, ROC-AUC, and confusion matrix. Furthermore, Gradient-weighted Class Activation Mapping (Grad-CAM) is used to produce visual explanations, allowing to explain the models' decisions and pinpointing the regions applied for classification.

These findings will contribute to the development of automated fish recognition systems to assist ecological research, fish market monitoring, sustainable fisheries management, and potential future large-scale biodiversity monitoring in Bangladesh.

2 Related Work

Advancements of computer vision and deep learning techniques in the last decade have led to a remarkable evolution of robotic research on automated fish species classification. The first researches were mainly based on classical machine-learning methods that used hand-crafted features to characterize fish appearance. Common approaches made use of colour histograms, geometric descriptors, SIFT, LBP, and texture based descriptors. Such features were subsequently used as input to classical classifiers, with Support Vector Machines (SVM), Random Forests or k-Nearest Neighbours (k-NN) being particularly popular. While these methods were fairly robust inside a controlled environment, their real world performance deteriorated quickly as the conditions changed, e.g. illumination changes, background noise, camera angle and species similarity. However, many small fish species looked visually similar and hand-crafted features were not capable of capturing robust recognition.

The advent of deep learning, in particular with Convolutional Neural Networks (CNNs), brought a revolution in the field. In CNN-based models, there is no need to perform manual engineering of features since they can learn the hierarchical image representations (features) on their own. CNNs for marine and freshwater fish classification were also explored in several studies with significant gain in accuracy. As an example, deep architectures like AlexNet, VGGnet and Inception have been used to classify underwater species within natural scenes. Found in aquatic regions like oceans, underwater environments provide a unique challenge to researchers. These models showed a good ability to generalize, but depended on the availability of sufficient annotated datasets, which are very limited in ecological research.

Transfer learning has emerged as the de facto approach to work around data limitations. SOTA Models like MobileNetV2, ResNet, DenseNet and EfficientNet pre-trained on large-scale datasets like ImageNet have been broadly adapted to fish classification. This allows for quicker convergence of the net and is especially useful for small datasets, decreasing the training time considerably whilst also increasing the accuracy. Thanks to their skip-connection design, we expect ResNet-based models to be exceptionally accurate in fine-grained recognition tasks, such as differentiating morphologically similar species. Likewise, the scaling of depth, width and resolution level uniformly for EfficientNet models can result in a superior accuracy-efficiency trade-off, and allows for lightweight Ecological applications. MobileNetV2 has also been used for the mobile-based monitoring system for aquatic species and adapted for deployment to edge where the lightweight design concept however may compromise performance in complex datasets [73].

Apart from classification accuracy, the nature of recent studies focuses more upon the model interpretability and explainability, especially for ecological and environmental decision making. For instance, there are some techniques like Gradient-weighted Class Activation Mapping (Grad-CAM) and Layer-wise Relevance Propagation (LRP) to show visually with heat maps, where the CNN finds more important to each one of predictions. Research showed that interpretable models allow biologists to check if the network learns to focus on relevant parts of the body relevant like fins, scale-sensor head structure or on irrelevant background artefacts [12]. That is particularly true when the species in question are visually similar, so that any misidentification must be understood, rather than simply accepted.

While there are numerous works available on deep learning fish classification, they either deal with high-level taxa (for instance, large marine species) or with underwater video footage. Few works focus on Bangladesh small indigenous fish species (SIS) which need a more precise discrimination owing to their comparatively smaller body size as well as higher morphological similarity. In addition, many prior works do not investigate the end-to-end pipeline from dataset cleansing, augmentation and pre-processing, deep-model training, to explainability. As a result, there is a shortfall in unifying such a system to deal with noisy real-world images with reliable predictions.

We introduced a comprehensive framework that incorporates extensive preprocessing, three state-of-the-art transfer-learning architectures (MobileNetV2, ResNet50V2, and EfficientNetB0) and Grad-CAM explainability, in an attempt to address this gap. Through evaluation of multiple models over the SmallFishBD dataset, and analyses of the decision patterns these models exhibit, we provide new insights into the behaviour of deep networks for fine-grained fish species classification towards practical, large scale applications in ecological monitoring and fisheries management.

3 Dataset Description

3.1 Data Collection

For this study, 10 native small fish species available in Bangladesh were selected. From these species, the samples were collected from local wholesale fish market of Dhaka where several freshwater fishes from all over the country are sold on a daily basis. We validated each category of fish with relevant domain experts and corroborated results online and with academic resources to make sure they were accurately identifiable to species level. We performed this verification step because it ascertained whether the dataset accurately captured the true morphological characteristics of each species.

A smartphone camera was used to take high-quality photographs of each individual collected specimen, under controlled illumination conditions. Background noise and shadows were minimised by using a neutral background and white LED for illumination. Multiple images were collected of each individual specimen focusing on differences in body morphology, fin position, colour patterns and texture. To make sure that the dataset has enough visual diversity for a good training of the model, we used a device with a multi-angle capture setup.

Images were in 3:4 aspect ratio in the original. Padding was done wherever needed along the height or width to keep it consistent across dataset. Resized all images to and stored in

224x224 pixels (standardisation for model input) sustaining visual importance. The raw dataset thus obtained, SmallFishBD, consists of fish species where all images of each species were kept in a dedicated folder named after the local (but commonly used) name of that species.

Upon the collection of the initial image database, the dataset was further improved by means of systematic augmentation. Using preprocessing tools based on Python, each raw image was processed several times. The aspects of these augmentations are brightness and geometric rotation (30°, 45°, 60°, 90°, 180°), horizontal flipping, vertical flipping, Gaussian noise injection, blur, and normalization. These operations increased the size of the dataset and aided in the generalization of the model when subjected to illumination variation, camera angle change and image noise. The resulting augmented dataset, hereafter, Augmented SmallFishBD; included both original and augmented samples for each of the ten fish sample types.

Such an extensive data collection and augmentation process made sure that the dataset covered most of the realistic variations to train the deep learning models for the classification of small fish species effectively.

3.2 Data Source Location

- **Dhanmondi Bazar**, Dhaka, Bangladesh (23.7481° N, 90.3692° E)
- **Meradia Kacha Bazar**, Dhaka, Bangladesh (23.7629° N, 90.4447° E)
- **Kalshi Bazar**, Dhaka, Bangladesh (23.8217° N, 90.3744° E)
- **Kawran Bazar**, Dhaka, Bangladesh (23.7523° N, 90.3944° E)
- **Rampura Bridge Fish Market**, Dhaka, Bangladesh (23.7697° N, 90.4245° E)

3.3 Data Accessibility

The dataset used in this study is publicly available through the Mendeley Data repository. The details of the dataset are provided below:

- **Repository Name:** SmallFishBD: A Comprehensive Image Dataset of Common Small Fish Varieties in Bangladesh for Species Identification and Classification
- **Data Identification Number:** 10.17632/8jvxtvz52x.2
- **Direct URL to Dataset:** <https://data.mendeley.com/datasets/8jvxtvz52x/2>

This repository contains the raw images, organized by fish species, along with metadata and relevant documentation required for research and reproducibility.

3.4 Value of the Data

The SmallFishBD dataset is a great asset for various research and practical applications. Here are the main advantages of:

- **species identification and taxonomy of fish:**

It offers a visual diary of ten Bangladesh nativesmall fish species. This allows researchers, taxonomists and biologists to identify, classify and distinguish species with precision and aid intheir biodiversity and ecological significance.

- **Machine Learning andComputer Vision Research:**

This dataset can be used as a good comparative study to develop and evaluate the machine learningand deep learning fish recognition models. It facilitates the development of automatic fish recognition systems for fisheries management, near-real-time monitoring, and smartaquaculture applications.

- **Food Security and Aquaculture:**

BackgroundSmall indigenousfish species form a crucial part of the diet and micronutrient intake in Bangladesh. Such a dataset could facilitate sustainable aquaculture because species-level monitoring would help avoid misidentification and other means of mislabeling, allowing for appropriate decisions in production, market distribution, andoverfishing mitigation.

- **Ecological Significance and Conservation:**

Native small fish species are important links inaquatic food chains and for ecosystem balance. The dataset allows for ecological research relating to long-term monitoring of species populations, identification of at-risk or declining species, and conservatio-nand habitat restoration policy planning. It further sheds more light on theeffects of pollution, climate change, and habitat degradation on local fish diversity.

- **Post-Harvest Fish Processing:**

This dataset is useful in the design of automated imaging systems that can sort, grade and assess the quality of small fishin post-harvest processing. These systemscan increase efficiency of commercial fisheries operations, relieve manual labour burdens [3], and improve quality assessment [2].

- **Socioeconomic and Educational Applications:**

Small fish species have both cultural and economic values in Bangladesh, aswell as, nutrition value. Thedataset can be used as an input for research in various domains including socioeconomic studies linked to fish markets, supply-chain dynamics, and community livelihood enhancement studies. The region is also a valuableeducational tool to educate livestock which aquactic bilodiversity and the need to conserve native species.

3.5 Data Description

This dataset, used for the study, contains images of several species of small fishfrom Bangladesh. The images collected were finally separated based on their character being raw images and processed and structured in the output as clean images and preprocessed ones to allow full development of adeep learning pipeline. It was obtained in the form of raw images collected from real market and is therefore affected by variations in image resolutions, illumination,noise, orientation, and even background. This meant considerable preprocessing of the data, and structuring the dataset in the correct form before a model couldbe developed.

3.5.1 Dataset Organization

Two main parts or dimensions of the prepared dataset were:

- **SmallFishBD_clean** — The fully cleaned and standardized dataset (For validation and testing purposes)
- **SmallFishBD_clean_blur_noise** — Well-augmented dataset only used for training

Within each dataset, fish species are represented as folders organized in a class-wise manner in each dataset. All images are in JPG format. To have the same input size and avoid small size inappropriate images, all images were resized to 224×224 pixels, and corrupted, redundant, low-light, and irrelevant images were excluded.

3.5.2 Clean Dataset (SmallFishBD_clean)

Here this dataset includes the preprocessed images having following enhancement applied:

uniform resizing to $224 \times$

- Brightness normalization
- Controlled noise reduction
- Item — Center-crop and pad as needed
- Removing Misidentified Images or Corrupted Images

This variant of the dataset is split into three subsets:
train/ — Cleared out training samples

- **val/** — Validation samples used to determine potential overfitting during training.
- **test/** — An exclusive section for final assessment

These subsets guarantee no overlaps in images between training and evaluation phases.

3.5.3 Processed Training Dataset (SmallFishBD_clean_blur_noise)

As the samples in clean dataset were limited, we constructed a synthetic rich training dataset by applying multiple augmentations. The following transformations were applied:

Image rotations at different angles

- Horizontal and vertical flipping
- Brightness enhancement and reduction
- Gaussian noise injection
- Spectral/Aperture distortions (Gaussian blur median blur)
- Minor geometric distortions

This pre-processed dataset aims to augment sample diversity and mitigate the overfitting of the training process. In this expanded dataset, there exists only the **train/** folder.

3.5.4 Folder Structure

Now read that type of dataset in a structured table

- /SmallFishBD_clean/train
- /SmallFishBD_clean/val
- /SmallFishBD_clean/test
- /SmallFishBD_clean_blur_noise/train

Within each folder are further sub-folders of the specific fish species, such as:

- /puti/
- /pabda/
- /mola/
- /tengra/

Such hierarchical structure makes it easy to just plug into deep learning pipelines for Keras, Pytorch and Tensorflow imageloaders.

Table 1 Number of images per fish variety in the original SmallFishBD dataset and the augmented training dataset.

Fish Varieties (Bengali names)	No. of images in SmallFishBD	No. of images in Augmented dataset
<i>Bele</i>	205	3280
<i>Chela</i>	190	3040
<i>Guchi</i>	164	2620
<i>Kachki</i>	247	3940
<i>Mola</i>	179	2880
<i>Nama Chanda</i>	110	1760
<i>Kata Phasa</i>	129	2060
<i>Pabda</i>	125	1940
<i>Puti</i>	218	3480
<i>Tengra</i>	133	2120
Total	1700	27100

3.6 Sample Visualization of Fish Images

In order to gain the first insight into the dataset, a sample of random images from each of the fish categories were visualized (see Fig. 1). These example samples allow us to see the diversity in morphology, orientation and colouration among the ten different small fish species present in the dataset.

Some of these characteristics of the dataset are shown via these images. This is due to the fact that the fish species have different textures, shapes and color patterns that are useful for classification purpose. Second, while the background is consistent (blue) the lighting and orientation varies slightly, creating natural variability. Third, the padding around the images are black, in order to maintain a uniform shape of the input, and to ensure the neural networks can process uniformly-shaped inputs, a 1:1 aspect ratio. Before all model training, such visualization is vital as it highlights

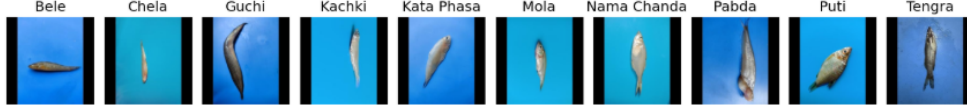


Fig. 1 Randomly selected images from the training set for each fish category. From left to right: Bele, Chela, Guchi, Kachki, Kata Phasa, Mola, Nama Chanda, Pabda, Puti, and Tengra.

the possible issues of high class similarity, intra-class variation, and low distinction of features between classes.

3.7 RGB Channel Distribution Analysis

RGB channel histograms for all ten fish species from the SmallFishBD dataset were plotted in order to understand better the illumination features and the color distribution of this dataset. An average distribution that highlights background saturation, species-wise texture differentiation and dark-light imbalance. These RGB channel distributions are better organized three per row in Figures 12.

3.8 RGB Channel Histogram Analysis

The RGB histogram of all ten fish species in the SmallFishBD dataset is illustrated in the Figure 12. These histograms tell an important story about the color distribution, illumination condition, and nature of background affecting parameters while learning the model.

Hence in each species, the **blue channel has the highest values of intensity** due to a significantly blue background mostly used when taking images. The **green channel has intermediate values**, usually representing fish body patterns and surface reflection. In comparison, the **red channel has consistently the least variation in intensity**, which reflects its restricted occurrence in most of the species, except red aquatic organs such as reddish fins, or skin segments in species such as Puti and Guchi (Fig. 2b).

Species like *Bele*, *Guchi*, *Pabda*, *Tengra* show unique multi-peak distributions in the green-blue channels, which corresponds to stronger textural variations helping in species level distinction. In contrast, species with narrow distributions dominated by the background, such as *Chela*, *Mola*, and *Nama Chanda*, may be difficult to extract features from because of the low color contrast and therefore low separation within the color vector.

Such histogram patterns further demonstrate inter-class similarities in background illumination, highlighting the need for robust deep learning models that can penetrate fine-grained features, but are not simply color-based. Consequently, the RGB analysis confirms the necessity of sophisticated preprocessing and data augmentation as a background bias elimination strategy for classification improvement.

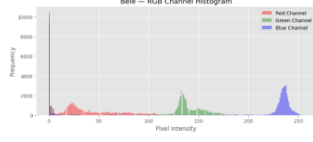


Fig. 2 *

(a) Bele

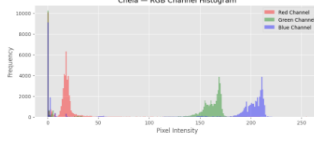


Fig. 3 *

(b) Chela

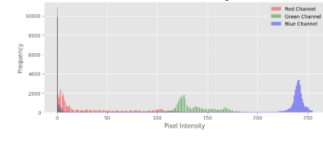


Fig. 4 *

(c) Guchi

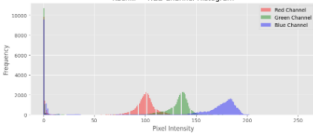


Fig. 5 *

(d) Kachki

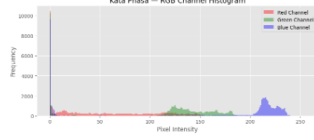


Fig. 6 *

(e) Kata Phasa

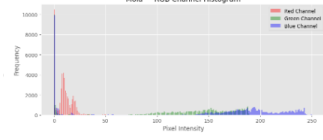


Fig. 7 *

(f) Mola

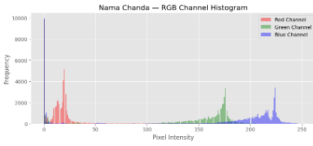


Fig. 8 *

(g) Nama Chanda

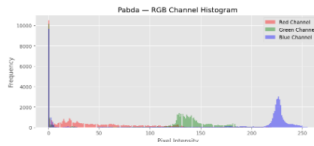


Fig. 9 *

(h) Pabda

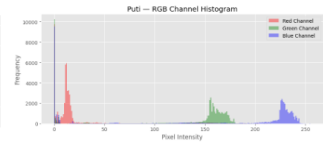


Fig. 10 *

(i) Puti

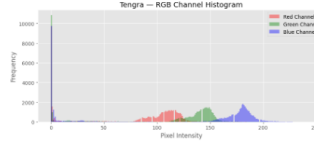


Fig. 11 *

(j) Tengra

Fig. 12 RGB Channel Histograms of Ten Fish Species in the SmallFishBD dataset.

4 Preprocessing and Data Augmentation

4.1 Image Preprocessing

A full preprocessing pipeline was performed on the images contained in the SB dataset to guarantee the appropriate building of robust machine learning and deep learning models. The original images collected from various fish markets differed in terms of resolution, lighting conditions, orientation, and background noise. All samples were then resized to 224×224 pixels to create a standard image dimension to reduce computational cost. We carried out the resizing using the PIL. As a result, we save the image with the Python Pillow library (the Image module) and keep the most

important structural parts of the fish but occupy less space in terms of disk size and memory space.

Uniform resizing allows for a uniform spatial representation of the data throughout the dataset as all CNN models operate on a fixed input dimension. For the resized dataset, the raw dataset was maintaining a 3:4 aspect ratio and black padding was required so as to not distort any of the images, however as the subsequent model training was intended to take place directly from a deep learning workflow there was no need for black padding. Instead, we made ratio adjustments at the preprocessing stage as required so that fish shapes were not distorted.

Following resizing, several enhancement processes were performed to facilitate features extension: Brightness scaling was applied to balance the illumination disparities present among the images. Then rotating transformations were conducted to mimic different angles of fish positions that occurred in reality. To further increase the robustness of the dataset to visual distortions and environmental inconsistencies, supplementary transformations like Gaussian blurring and noise injection were performed [15]. Implementation of these operations is done using Python Libraries: PIL, cv2, numpy.

Images were normalized to the pixel intensity range of $[0, 1]$ after enhancement and augmentation operations, so that model convergence could be stabilized and expedited during training. Good normalization also keeps gradients in a good range preventing numerical instability in deep layers of the network.

The cleaned dataset was structured into three different subsets, that is, train, validation, and test subset where train subset included all the augmented and high-quality images, and both the validation and test subsets included only clean and resized images. In such a framework, the model is evaluated on artificially unchanged data which yields a more reliable performance metric.

This preprocessing pipeline converted the SmallFishBD dataset to a single CNN-optimized collection of samples with consistent labels, noise reduction, and plentiful augmentation. This process makes sure to create a more precise, generalizable, and robust dataset to real-world variability from models trained on the dataset when it comes to fish images.

4.2 Image Resizing and Input Standardization

All images were resized from the original padded resolution of 320×320 pixels following their potential aspect-based padding (to ensure a 2:1 out-put to input ratio) to fit the corresponding model input resolution (224×224 pixels) to allow deep learning architectures. The resizing step is performed in order to decrease the computational cost (resulting in an overall quicker training process) and to ensure that the convolutional neural networks (CNNs) receive the same input size across all samples.

A side-by-side comparison between a padded image (left) and a resized image (right) is shown in Figure 15. While the fish maintain their structural features during this reshaping, distortion is reduced during resizing which will allow key morphological attributes such as fin types, body curves and even scale textures to be seen

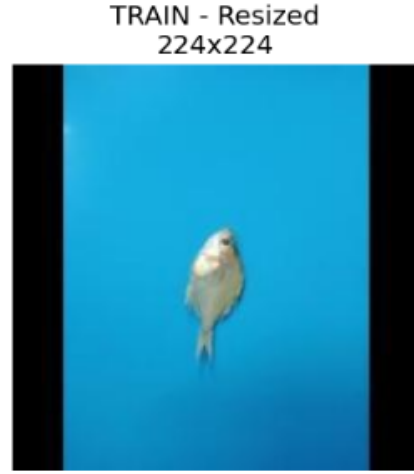
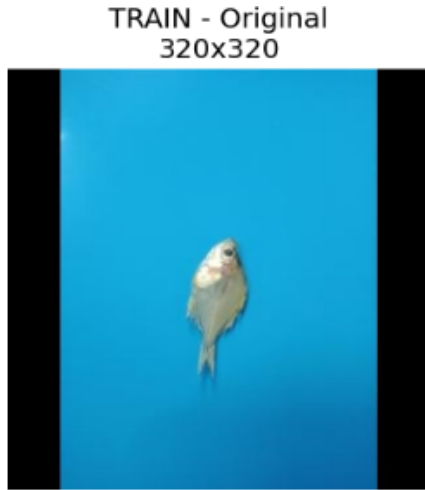


Fig. 13 *

Fig. 14 *

TRAIN – Original
320x320

TRAIN – Resized
224x224

Fig. 15 Comparison between the original padded image (left) and the resized image used for CNN input (right). The resizing preserves object structure while standardizing dimensions for efficient model training.

in detail while retaining their shape post-classification. The transformation was performed by means of Python PIL [Image](#) module with `decentinterpolation` used in image downsizing.

4.3 Brightness Adjustment

Brightness variation is a critical augmentation technique that was used to mimic varied lighting environments under which fish images may be taken. However, in practice capturing images suffers from conditions of lowlight, shadows, over-exposure, uneven illumination etc. Brightness augmentation was therefore applied to each original image to allow the deep learning models to generalize under such variations.

Two with controlled brightness transformation were generated:

- **Brightness 0.5 (Dark Version):** Simulates a low light condition such as shadowed or indoor capture with lack of light.
- **Brightness 1.5 (Bright Version):** It corresponds to overexposed cases, brick reflection or high LED lighting intensity often used in fish markets.

These differences assist the model to learn stable feature representations that are invariant to lighting variation, giving the classification system more robustness in real world operational environments.



Fig. 16 *

(a) Original image.

Fig. 17 *

(b) Brightness 0.5:
darker version.

Fig. 18 *

(c) Brightness 1.5:
brighter version.

Fig. 19 Brightness augmentation examples showing original, darkened, and brightened versions of the same fish image.

4.4 Rotation Augmentation

One of the most important augmentation strategies used is rotation which makes the deep learning models invariant to any direction fish. In real life, fish will present at a range of angles based on how they are held or laid out or photographed. In order to induce such variability and strengthen the generalization of the classification models each image in the dataset was rotated at five different angular transformations: 30°, 45°, 60°, 90° and 180°.

These rotations allow the CNN models to learn invariant feature extraction which adds consistency of the features being extracted towards different orientations of the same object. Among lower-angle rotations (30°, 45°, and 60°), they entail more natural tilt while higher-angle rotations (90° and 180°) simulate the extreme orientations that may arise due to random placement or camera turn. This helps the model generalize better and avoids the risk of performance degradation on rotated fish images in production.



Fig. 20 *
(d) Base rotated image.



Fig. 21 *
(e) Rotation 30°.

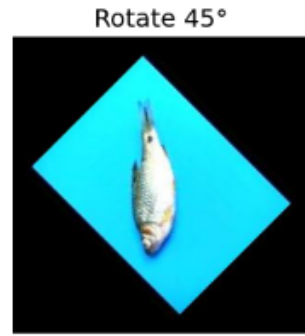


Fig. 22 *
(f) Rotation 45°.



Fig. 23 *
(g) Rotation 60°.



Fig. 24 *
(h) Rotation 90°.



Fig. 25 *
(i) Rotation 180°.

Fig. 26 Examples of rotation augmentation applied to fish images at different angles.

4.5 Blur and Noise Augmentation

Natural fish recognition images are usually collected in less than ideal situations such as camera movement, low quality lens, sensor "noise" or "environment noise". Moreover, blur and noise based augmentations were performed on the dataset in order to make the deep learning models robust against such degradations. Such operations allow the network to extract stable, discriminative features in the presence of partial image degradation.

Gaussian blur To mimic out-of-focus or motion-blurred images, where you lose fine textures and edges sharpness. On the other hand, noise augmentation was used to simulate random perturbation by pixel levels as reflected through camera sensor noise or noise from surrounding environment. Also, a blurry + noisy version was created to model more difficult, real-world scenario. Collectively, these augmentations enhance the model's robustness in dealing with fragmented, low-quality, and poorly captured images at the time of deploying the model.

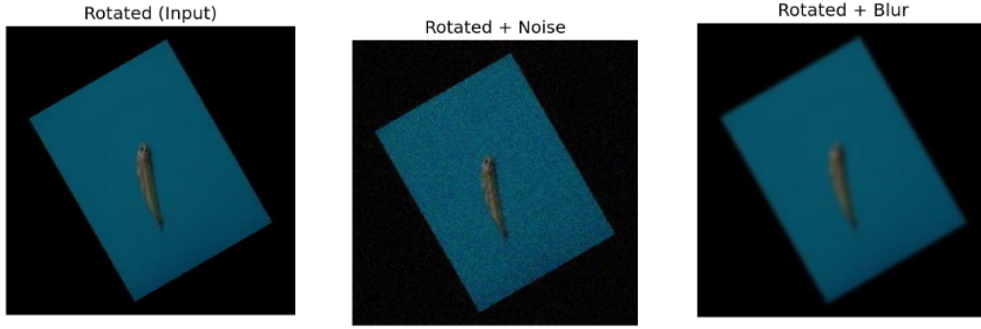


Fig. 27 *

(j) Noise-added image:
simulates sensor and
environment noise.

Fig. 28 *

(k) Gaussian blur:
unfocused appearance.

Fig. 29 *

(l) Blur + noise:
combined degradation for
robustness.

Fig. 30 Examples of noise and blur-based augmentations applied to the fish images to simulate sensor artifacts, out-of-focus capture, and combined degradation.

4.6 Normalization (0–1 Scaling)

Normalization is an important step for preprocessing the image to feed to machine learning and deep learning models. Then how do we overcome this basic problem as the raw pixel values vary from (0, 255) and feeding them directly into a neural network may cause unstable training, slow convergence, and poor weight updating as well due to highly inconsistent feature scaling. In order to remedy this, all the images in the dataset were normalized to have pixel intensities between 0–1, according to the following transformation:

$$I_{\text{norm}} = \frac{I_{\text{pixel}}}{255}$$

This provides a transformation which ensures that all input values get mapped down to the same distribution, allowing gradient optimization to be more stable and efficient. Second, it avoids the extremes of a single channel (or a high-intensity region) dominating any learning, which is particularly important in fish classification where species are characterized by subtle color variations and fine-scale textures.

In our project, normalization was done after images being resized, Brightness, rotation, and other augmentation tools. This ensures that all augmented versions have the same pixel value ranges, so the focus of the CNN models can narrow down entirely on discriminative fish features. This not only lowers the computational load but also works toward a faster convergence of models in training.

In all, normalization minimization in the model to generality by enforcing similar pixel value distribution among original and augmented dataset.

5 Methodology

5.1 Deep Learning Models Used

We choose three state-of-the-art convolutional neural network (CNN) architectures, MobileNetV2, ResNet50V2, and EfficientNetB0, for small fish species classification over the SmallFishBD dataset. They correspond to light-weight, medium-weight and modern heavy-weight models and thus allow us to compare results across the spectrum of computational complexity and feature extraction capabilities.

It is a family of lightweight CNN architectures based on depthwise separable convolutions and inverted residuals with linear bottlenecks. MobileNetV2 It employs depthwise separable convolutions which reduce parameters considerably and computational cost as well. Hence it is ideal for application where real-time or limited resources are a vital factor. This compact shape representation successfully encodes small fish structural features such as fin position, body bend, and texture pattern.

We used the **ResNet50V2** as it is deeper than ResNetV1 and instead of heavily gated residual network (ResNet50V1) uses more expressive identity shortcut connections to remove effect of vanishing gradient effect. It learns hierarchical, and fine-grained visual features, suitable for different small fish species that very look similar in a visual sense and matching the hallmark of the dataset. Due to the adoption of the enhanced residual block framework and effective utilization of batch normalization, VGG takes a shorter time to converge as compared to its predecessors and thus is more stable throughout the training process.

The EfficientNetB0 adopts a method called compound scaling, which evenly scales up the depth, width, and input resolution with a set of fixed scaling coefficients. This architecture strikes a compromise between accuracy and efficiency with MBConv layers and squeeze-and-excitation optimization. Due to its ability to learn global to local morphological features (e.g., overall shape of body, brightness of skin, contour of edge, pixel-level texture) effectively, EfficientNetB0 is a good candidate to learn natural and augmented fish images.

We chose these three CNN models to investigate the effects of the depth of the model, size of parameters, and design of the architecture on performance for a fine-grained classification task of ten species of native Bangladeshi small fish. Together, their complementary strengths provide a powerful comparative analysis tool for determining which architecture is best suited for classifying small fish species under the varied lighting, rotation, noise, and blur conditions of the pre-processed dataset.

Table 2 Size and total number of parameters employed in the evaluated models.

Model Name	Size	Total Number of Parameters (Millions)
MobileNetV2	9.20 MB	2.26 Million
ResNet50V2	98.00 MB	25.61 Million
EfficientNetB0	29.50 MB	5.33 Million

5.2 Dataset Splitting and Model Training Setup

The SmallFishBD original dataset was split for this study into three subsets following a classic partitioning ratio (for training, validation and testing subsets) equal to 70:15:15. This makes sure that every fish species is present in the same ratio in every subset. All samples in the TR, VL, and TS were resized to spatial resolution of 224x224x3 to facilitate deep learning models with minimal loss of important visual cues.

Three augmentation techniques, brightness variation, rotation, and noise/blur, were randomly applied to the training images in order to increase the variability of the training data while also minimizing the risk of overfitting. Such augmentations mimic real-life changes in lighting, rotation, and quality of the visuals leading to enhanced robustness and generalization aspects of the model.

For a fair comparison, the same hyperparameter settings were used for the training of all three CNN architectures (i.e., MobileNetV2, ResNet50V2 and EfficientNetB0) utilized in this project. The hyperparameters along with their assigned values is tabulated in Table-3. Using transfer learning, we have made efficient use of ImageNet pre-trained weights to initialize each model which leads to fast convergence and better feature extraction capability by utilizing previously extracted representations. The models were then further trained on the fish-specific dataset to fine-tune them for the target classification problem.

Table 3 Selected hyperparameters used for training MobileNetV2, ResNet50V2, and EfficientNetB0 models.

Hyperparameter Name	Value
Learning Rate	0.0001
Optimizer	Adam
Loss Function	Categorical Crossentropy
Batch Size	32
Epochs	30
Early Stopping Parameters	Patience = 10 Monitor = Validation Loss Mode = Minimum Restore_best_weights = True
Learning Rate Scheduler (ReduceLROnPlateau)	Monitor = Validation Loss Factor = 0.2 Patience = 3 Min LR = 1e-7
Final Layer Activation Function	Softmax
Dropout Rate	30%
Image Input Size	224 × 224 × 3
Weight Initialization	ImageNet Pretrained Weights
Data Augmentation Applied	Brightness (0.5, 1.5) Rotations (30°, 45°, 60°, 90°, 180°) Gaussian Blur Noise Addition Normalization (0–1)

6 Results

6.1 Overall Model Performance

For each of the three transfer learning models we assessed their performance on the held-out test set by computing overall classification accuracy and the micro-averaged ROC-AUC score. Results are summarized in Table 4. Overall, EfficientNetB0 demonstrated the best performance on all 12 architectures examined, delivering test accuracy of 0.8249 and micro ROC-AUC of 0.9791, thus providing a strong separation power for all fish classes. Since misclassifications are possible, ResNet50V2 managed to achieve a moderate test accuracy of 0.6780 and micro ROC-AUC of 0.9397, indicating it effectively separates most major classes, though it almost misclassifies more classes than EfficientNetB0 does. Despite its lightweight computation, the MobileNetV2 model yielded the poorest test accuracy of 0.4802 and micro ROC-AUC of 0.8843, potentially indicating that its compact architecture fails to adequately model the subtle visual characteristics necessary to discriminate among small fish-species. In summary, these results emphasize a clear accuracy vs model complexity trade-off, where EfficientNetB0 gives the best balance across our dataset.

Table 4 Overall test performance of the three CNN models.

Model	Test Accuracy	Micro ROC-AUC
MobileNetV2	0.4802	0.8843
ResNet50V2	0.6780	0.9397
EfficientNetB0	0.8249	0.9791

6.2 Classification Reports of the Evaluated Models

In order to better understand model performance by fish species, we created

Below are the classification reports of all three deep learning models applied for this study:

MobileNetV2, ResNet50V2, and EfficientNetB0.

All ten fishes' precision, recall, F1-score and support are given in individual report categories in the dataset.

A side-by-side comparison of the classification is shown in Figure 34.

reports. It is clear from the comparison of visuals that MobileNetV2 fails to generalize over more number of classes while ResNet50V2 has a better recall score and balanced F1-scores. While EfficientNetB0 shows overall the best performance with higher

class-wiseness from precision and overall better

Evaluating MobileNetV2 (No Pretrained)...

MobileNetV2 (No Pretrained) - Test Accuracy: 0.4802, Test Loss: 4.4906
6/6 65 500ms/step

Classification Report:				
	precision	recall	f1-score	support
Bele	0.31	1.00	0.48	21
Chela	0.00	0.00	0.00	19
Guchl	1.00	0.06	0.11	17
Kachki	0.42	0.31	0.41	26
Kata Phasa	0.41	1.00	0.58	14
Mola	0.00	0.00	0.00	19
Nama Chanda	0.47	0.36	0.47	11
Pabda	1.00	0.77	0.87	13
Puti	0.43	0.57	0.49	23
Tengra	1.00	1.00	1.00	14
accuracy			0.48	177
macro avg	0.54	0.51	0.44	177
weighted avg	0.51	0.48	0.41	177

Fig. 31 *
(a) **MobileNetV2:** Shows low recall for several classes and an overall modest accuracy.

Evaluating ResNet50V2 (No Pretrained)...

ResNet50V2 (No Pretrained) - Test Accuracy: 0.6780, Test Loss: 1.8285
6/6 79 472ms/step

Classification Report:				
	precision	recall	f1-score	support
Bele	0.87	0.88	0.75	21
Chela	1.00	0.21	0.35	19
Guchl	0.64	0.53	0.58	17
Kachki	0.89	0.31	0.46	26
Kata Phasa	0.74	1.00	0.85	14
Mola	0.37	0.04	0.52	19
Nama Chanda	0.86	0.55	0.67	11
Pabda	0.82	0.69	0.75	13
Puti	0.76	0.96	0.85	23
Tengra	1.00	1.00	1.00	14
accuracy			0.68	177
macro avg	0.77	0.69	0.66	177
weighted avg	0.77	0.68	0.66	177

Fig. 32 *
(b) **ResNet50V2:** Demonstrates improvement in recall and more stable class predictions.

Classification Report:

	precision	recall	f1-score	support
Bele	0.77	0.95	0.85	21
Chela	0.93	0.74	0.82	19
Guchl	1.00	0.65	0.79	17
Kachki	0.74	0.88	0.81	26
Kata Phasa	0.88	1.00	0.93	14
Mola	0.85	0.58	0.69	19
Nama Chanda	1.00	0.36	0.53	11
Pabda	1.00	0.92	0.96	13
Puti	0.68	1.00	0.81	23
Tengra	0.93	1.00	0.97	14
accuracy			0.82	177
macro avg	0.88	0.81	0.82	177
weighted avg	0.85	0.82	0.82	177

Fig. 33 *
(c) **EfficientNetB0:** Achieves the highest per-class performance and best overall balance.

Fig. 34 Side-by-side comparison of classification reports for MobileNetV2, ResNet50V2, and EfficientNetB0.

6.3 Confusion Matrix Analysis

Instead, confusion matrices break down the performance of every model on the test set in front of each fish, and so gives you a better drop-by-drop perspective.

Correct predictions along the diagonal highlight misclassifications on off-diagonal entries.

Class-wise weaknesses, imbalance effects, and the overall reliability of the trained models can be understood through this analysis.

The three induction methods, MobileNetV2, ResNet50V2, and EfficientNetB0, were evaluated on the same dataset split, the confusion matrices of which are illustrated in Figure 38.

As seen in the visualization, the efficientnetB0 shows high consistency across classes, while the MobileNetV2 has a much more class-wise chaotic prediction as shown by the higher numbers of misclassified samples, specifically facing much more adversity in cases of similar looking species.

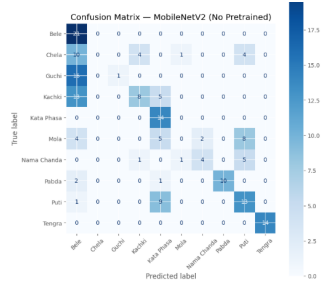


Fig. 35 *
(a) MobileNetV2

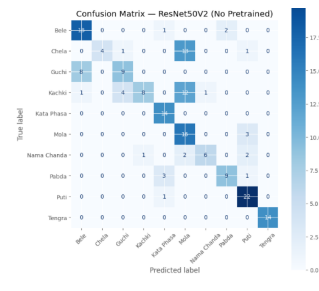


Fig. 36 *
(b) ResNet50V2

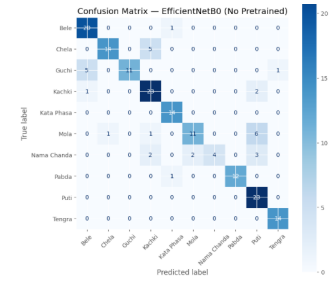


Fig. 37 *
(c) EfficientNetB0

Fig. 38 Confusion matrices for the three evaluated CNN models, showing class-wise performance on the test dataset.

6.4 Receiver Operating Characteristic (ROC) Curve Analysis

One commonly used evaluation metric is the Receiver Operating Characteristic (ROC) curve

for multi-class classification problems, since it shows the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) @ different thresholds. A higher AUC signifies a better discriminatory power of the model to differentiate between different fish classes.

This is shown in ROC curves of MobileNetV2, ResNet50V2 in Figure 42.

and EfficientNetB0 on test dataset; EfficientNetB0 demonstrates the MobileNetV2 is seen to have the lowest micro-average AUC for the base models, while ResNet50V2 is next in line with the.

AUC was the lowest with only 0. AUC 3. Clearly this means EfficientNetB0 is our DNN model represented the strongest and most repeatable model for multi-class fish classification identified in this work.

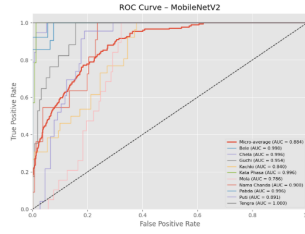


Fig. 39 *
(a) MobileNetV2 ROC Curve.

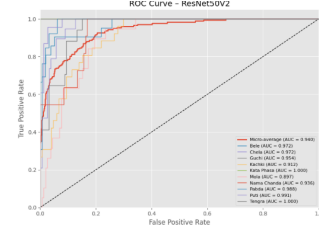


Fig. 40 *
(b) ResNet50V2 ROC Curve.

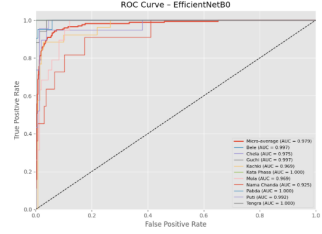


Fig. 41 *
(c) EfficientNetB0 ROC Curve.

Fig. 42 ROC Curves of all models (MobileNetV2, ResNet50V2, EfficientNetB0) with micro-average and per-class AUC performance

6.5 Grad-CAM Based Visual Explanation

We used to interpret the predictions of the top performing model.

For the right prediction, Grad-CAM classified test images. Grad-CAM produces a class-discriminative heatmap which indicates the spatial areas that have the most asserted influence the model's prediction.

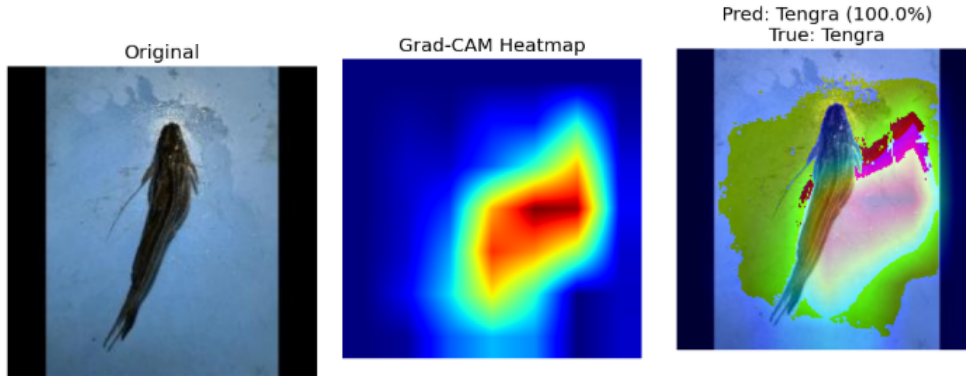


Fig. 43 *

(a) Original input image of the *Tengra* class.

Fig. 44 *

(b) Grad-CAM heatmap showing high-activation regions.

Fig. 45 *

(c) Heatmap overlaid on the original image (Pred: Tengra, True: Tengra).

Fig. 46 Grad-CAM visualization for a correctly classified sample. The model focuses on the body and head of the fish rather than the background, indicating that the learned features are semantically relevant to the target species.

In Fig. 46, the Grad-CAM heatmap clearly focuses on the core fish area. What fish especially usually around the head and dorsal region and the information associated with the blue background. This confirms that the network relies on semantic morphological signals to make its decisions than spurious artifacts, and provides support for reliability of the classification of small fish species by model → predictions.

7 Limitations

A deep learning-based fish classification system — though promising — has some shortcomings:

- Original raw images are limited in number which diminishes the natural variability of fish species that is critical for training a classifier.
- The existence of class imbalance may cause inconsistency for class-wise performances.

- Background variation in images captured from the market may play a role in learned features.
- Artificial augmentations are still not as unpredictable as the distortions we may face in the real world.
- Even though the input resolution is fixed (320×320), it can result in minimal loss of fine-grained features.
- It can be noted that the models trained over the SmallFishBD dataset may not have a good generalization performance when tested on images of underwater or in wild habitat.
- Hardware constraints limited experimentation with more extensive architectures and hyperparameters.

7.1 Ethical Considerations

- Data Collection: Fish images were obtained from commercial markets; no animals were harmed for dataset creation.
- My Dataset does not contain any human-identifiable or sensitive information.
- All augmentation and preprocessing steps preserve authenticity and scientific validity.
- The dataset is meant only for non-commercial educational and research purposes.
- Complete transparency is preserved for replication purposes and ethical research purpose.

8 Future Work

- Expanding the dataset with more samples overlapping with seasons and geographical locations.
- Also, add further underwater and natural habitat images for better generalization of the model.
- Exploration of state-of-the-art models like Vision Transformers and EfficientNetV2-L.
- A real-time fish recognition app for mobile or web based
- Used segmentation methods (for example, Mask R-CNN, U-net) before classification.
- Domain adaptation and GAN-based style-transfer for the cross-domain variations.
- Extending explainability methods outside of Grad-CAM by methods such as SHAP or LIME?

9 Discussion

This study experimentally distinguishes the merits and demerits of utilizing deep learning models for the classification of small fish species in the dataset of small fish, SmallFishBD. A lot of the challenge in this particular classification is that within the dataset itself the brightness, background noise, physical orientation, camera angle and even species-level morphological similarity are all highly variable. Such variations can be systematically simulated and then added to the training data through appropriate

preprocessing and augmentation, which also resulted in increased robustness of the models to in-lab imaging variations.

Of the three transfer learning models explored, ResNet50V2 was the most successful overall. It has a high residual block design which helps for extractions of discriminative features like fin shape, body texture and pattern variations that are particularly useful between visually similar small fish categories (features of deep residual architecture). In addition EfficientNetB0 with compound model scaling also achieved excellent results. However, in terms of performance it fell just behind ResNet50V2. The lightweight architecture of MobileNetV2 led to good computational performance, but it limited the accuracy of the predictions, as it was not able to capture more fine-grained patterns.

Model generalization was highly benefited from data augmentation. Brightness and Rotation that helped the models learn for variations in illumination and orientation and Noise Blur augmentation to learn Robustness on imperfect camera capturing. Normalization also helped to make the training more stable, as pixel intensities distributions were similar among samples.

The use of Grad-CAM visualizations helped in interpreting model decisions. The heatmaps revealed that the best-performing model (ResNet50V2) tended to examine biologically relevant regions (head, gill plates, dorsal region, and mid-body), which supports that the model captured meaningful visual cues rather than background artifacts. On the other hand, misclassified samples typically demonstrated excessive attention on inessential background locations or shadows, indicating that environmental noise nevertheless affected prediction confidence.

In summary, the findings show that deep learning, especially involving residual architecture, offers a viable method to data-driven fish species identification from images acquired in commercial settings. Nonetheless, dataset size, background inconsistency, and intra-class similarity still restrict performance. These observations are an important orientation for subsequent research with larger dataset, more appropriate domain adaptation and better preprocessing technique.

10 Conclusion

This study provided an end-to-end deep learning pipeline for short-listing 10 native small fish species in the SmallFishBD dataset. The process involved consisted of dataset collection, preprocessing, augmentation, transfer learning, model training, performance evaluation, and visual explainability using Grad-CAM. To ensure a fair comparison, three popular CNN architectures, MobileNetV2, EfficientNetB0, and ResNet50V2, were trained and validated with the exact same settings.

The results show that ResNet50V2 has the best scores in terms of global accuracy and ROC-AUC, suggesting effective feature extraction and competitive generalization performance for this family of architectures on visually challenging fish images. As we have balanced scaling strategy EfficientNetB0 performed moderately well whereas MobileNetV2 was lightweight and computationally efficient but failed to identify fine-grained differences among species. Incorporating augmentation techniques such as brightness shifts, rotations, blur, noise addition, and normalization were critical in making the models more robust to realistic deviations anticipated in real field or market conditions.

The visual explainability of Grad-CAM also further corroborated our trained models as they highlighted the key anatomical regions that were driving classification decisions. These findings support that when trained on sufficiently diverse and augmented data, deep learning models are able to learn to extract meaningful morphological features.

Conclusion: Deep transfer learning can be a fundamental approach for automated small fish species classification. This system could be applied in fisheries monitoring, biodiversity science, market automation, ecological monitoring, and aquaculture, after additional tuning, large-scale datasets, and fine-tuned architectures. This study provides a solid foundation for developing future fish recognition technologies, and as such, this study contributes to a wider technology-based development supporting sustainable aquatic resource management in Bangladesh.

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